

TRAPPES, France

WMO: 071450

Lat: 48.7744N

Lon: 2.0097E

Elev: 174

StdP: 99.25

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.6	-3.1	-8.8	1.8	-2.8	-6.5	2.2	-1.5	8.2	9.4	7.5	9.3	2.0	10	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	30.5	19.9	28.2	19.0	26.3	18.3	20.9	28.3	19.9	26.3	19.0	24.8	2.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	13.7	23.2	17.6	12.9	22.0	16.7	12.2	21.3	60.9	28.3	57.4	26.4	54.6	24.9	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.8	6.0	5.3	-6.9	34.6	2.5	2.5	-8.7	36.3	-10.1	37.8	-11.5	39.2	-13.3	41.0		
			-7.4	22.8	2.5	1.0	-9.2	23.5	-10.7	24.1	-12.1	24.6	-13.9	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	4.2	4.7	7.6	10.7	14.1	17.5	19.4	19.0	16.1	12.2	7.6	4.8
	DBStd	6.43	3.81	3.77	3.51	3.48	3.34	3.26	3.25	2.97	2.81	3.41	3.40	3.70
	HDD10.0	725	181	151	90	32	4	0	0	0	0	17	86	164
	HDD18.3	2638	438	382	333	231	139	55	26	26	79	189	322	419
	CDD10.0	1277	1	2	16	52	130	223	290	278	182	87	14	3
	CDD18.3	151	0	0	0	1	6	29	57	46	12	1	0	0
	CDH23.3	1311	0	0	0	11	53	243	507	399	92	5	0	0
	CDH26.7	383	0	0	0	0	4	66	162	136	16	0	0	0
Wind	WSAvg	2.5	2.9	2.9	2.8	2.6	2.4	2.4	2.3	2.1	2.1	2.3	2.5	2.9
Precipitation	PrecAvg	672	58	49	55	49	62	54	61	51	54	60	61	58
	PrecMax	853	120	106	110	112	118	159	152	114	127	164	99	118
	PrecMin	415	13	7	2	3	8	2	14	2	8	7	10	4
	PrecStd	113	29	27	28	27	30	31	33	27	33	42	22	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	15.5	20.0	24.8	27.0	31.4	33.7	34.6	28.9	23.6	16.8	13.8
		MCWB	11.5	10.0	11.6	15.6	19.0	20.9	20.8	21.4	18.7	17.0	14.4	12.0
	2%	DB	11.8	12.6	17.4	22.3	24.8	28.6	30.4	30.4	26.2	20.8	15.2	12.4
		MCWB	10.5	9.3	11.1	14.1	16.7	19.4	19.8	19.7	17.9	16.3	13.3	11.0
	5%	DB	10.6	11.2	15.3	19.3	22.7	26.0	28.2	27.6	23.6	18.6	13.7	11.3
		MCWB	9.4	9.1	10.3	12.6	15.8	18.3	19.0	18.7	17.0	15.1	12.0	10.0
	10%	DB	9.5	10.0	13.1	17.1	20.4	23.9	26.2	25.2	21.6	17.0	12.4	10.2
		MCWB	8.3	8.2	9.7	11.6	14.7	17.4	18.4	17.6	15.9	14.2	11.0	9.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.0	11.3	13.1	16.7	19.6	22.4	21.8	22.0	19.9	18.0	15.2	12.3
		MCDB	12.8	13.3	16.9	23.4	25.8	29.1	30.2	31.0	26.1	21.5	16.2	13.3
	2%	WB	10.7	10.2	12.0	14.8	18.0	20.4	20.7	20.5	18.6	16.6	13.5	11.2
		MCDB	11.5	11.8	15.6	20.7	23.1	26.3	28.5	27.9	23.9	19.5	14.8	12.2
	5%	WB	9.5	9.3	11.1	13.3	16.5	19.2	19.8	19.4	17.6	15.6	12.2	10.2
		MCDB	10.4	10.8	14.2	18.2	21.1	24.4	26.5	25.7	22.1	18.2	13.4	11.2
	10%	WB	8.4	8.4	10.1	12.0	15.3	18.0	18.9	18.5	16.8	14.7	11.1	9.1
		MCDB	9.3	9.9	12.9	16.1	19.4	23.1	24.9	23.8	20.6	16.7	12.3	10.1
Mean Daily Temperature Range	5% DB	MDBR	4.6	6.0	7.7	9.0	8.9	9.2	9.6	9.7	9.2	7.2	5.3	4.8
		MCDBR	5.0	7.8	11.2	12.5	12.1	12.6	13.0	13.9	12.4	9.1	6.2	5.1
	5% WB	MCWBR	4.5	5.4	6.2	6.1	5.7	5.6	4.8	5.4	5.3	5.2	4.7	4.4
		MCWBR	4.9	6.5	8.9	11.3	10.7	10.8	11.6	11.7	10.3	8.0	5.7	4.8
Clear-Sky Solar Irradiance	taub		0.330	0.351	0.385	0.403	0.409	0.411	0.410	0.400	0.374	0.368	0.340	0.324
		taud	2.394	2.350	2.249	2.217	2.235	2.266	2.265	2.314	2.375	2.381	2.401	2.367
	Ebn at Noon		722	791	819	843	852	851	846	838	825	760	701	671
		Edh at Noon	68	88	114	130	133	130	128	117	100	85	67	61
All-Sky Solar Radiation	RadAvg		0.92	1.70	2.87	4.36	5.13	5.73	5.43	4.71	3.65	2.15	1.11	0.78
	RadStd		0.10	0.23	0.36	0.49	0.48	0.54	0.44	0.45	0.29	0.18	0.11	0.11

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page